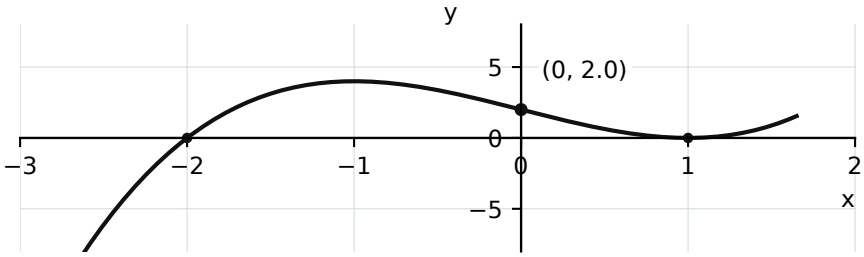
MATH 101 **94806**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-1)(x+3)}{(x-5)(x-1)}$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Find all zeros. $p(x) = x^3 - x^2 - 3x + 2$	2. 20 pts
3. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	3. (a) 16 pts (b) 2 pts (c) 2 pts
4. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = (x+1)^2 - 4$	4. (a) 6 pts (b) 4 pts (c) 10 pts

Test continues on the other side of the sheet.

MATH 101 **94806**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = \sqrt{x+1} - 2$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

94806

TEST PAGE 1

1. (a)
$\mathbb{R} \setminus \{1, 5\}$ 4 pts

(b)
$(1, -1)$ 4 pts

(c)
$x = 5$ 3 pts

(d)
$y = 1$ 3 pts

(e)
$(-3, 0)$ 3 pts

(f)
$(0, -\frac{3}{5})$ 3 pts

2.
$x = 2, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 20 pts

3. (a)
$1(x+2)(x-1)^2$
16 pts

(b)
$x \to -\infty : f(x) \to -\infty$
2 pts

(c)
$x \to \infty : f(x) \to \infty$
2 pts

4. (a)
$(-1, -4)$ 6 pts

(b)
-4 4 pts

(c)
$x = -3, 1$ 10 pts

TEST PAGE 2

5. (a)
$[-1, \infty)$ 4 pts

(b)
$[-2, \infty)$ 4 pts

(c)
$(3, 0)$ 3 pts

(d)
$(0, -1)$ 3 pts

(e)
$x \to \infty : f(x) \to \infty$ 4 pts

(f)
$x \to -1^+ : f(x) \to -2$ 2 pts

MATH 101 **56952**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 + 4x^2 + 4x + 1$	1. 20 pts
2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = \sqrt{x+4} - 2$	2. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-1)(x+3)}{(x-5)(x-1)}$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts

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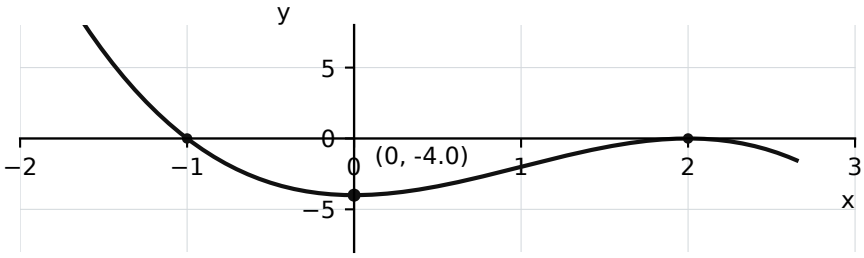
MATH 101 **56952**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	4. (a) 16 pts (b) 2 pts (c) 2 pts
5. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = (x+1)^2 - 4$	5. (a) 6 pts (b) 4 pts (c) 10 pts

56952

TEST PAGE 1

1.
$x = -1,\ -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 20 pts

2. (a)
$[-4, \infty)$ 4 pts

(b)
$[-2, \infty)$ 4 pts

(c)
$(0, 0)$ 3 pts

(d)
$(0, 0)$ 3 pts

(e)
$x \to \infty : f(x) \to \infty$ 4 pts

(f)
$x \to -4^{+} : f(x) \to -2$ 2 pts

3. (a)
$\mathbb{R} \setminus \{1, 5\}$ 4 pts

(b)
$(1, -1)$ 4 pts

(c)
$x = 5$ 3 pts

(d)
$y = 1$ 3 pts

(e)
$(-3, 0)$ 3 pts

(f)
$(0, -\frac{3}{5})$ 3 pts

TEST PAGE 2

4. (a)
$-1(x+1)(x-2)^2$ 16 pts

(b)
$x \to -\infty : f(x) \to \infty$ 2 pts

(c)
$x \to \infty : f(x) \to -\infty$ 2 pts

5. (a)
$(-1, -4)$ 6 pts

(b)
-4 4 pts

(c)
$x = -3, 1$ 10 pts

MATH 101 **55300**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 - 4x + 3$	1. 20 pts
2. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-1)(x+3)}{(x-5)(x-1)}$	2. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
3. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = \sqrt{x+9} - 2$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **55300**

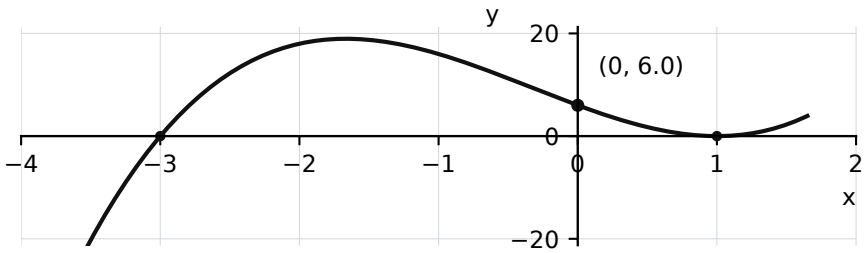
Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end.



4. (a)

16 pts

(b)

2 pts

(c)

2 pts

5. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros.

$f(x) = (x + 1)^2 - 4$

5. (a)

6 pts

(b)

4 pts

(c)

10 pts

55300

TEST PAGE 1

1.

$x = 1,\ -\frac{1}{2} + \frac{\sqrt{13}}{2},\ -\frac{\sqrt{13}}{2} - \frac{1}{2}$ 20 pts

2. (a)

$\mathbb{R} \setminus \{1, 5\}$ 4 pts

(b)

$(1, -1)$ 4 pts

(c)

$x = 5$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-3, 0)$ 3 pts

(f)

$(0, -\frac{3}{5})$ 3 pts

3. (a)

$[-9, \infty)$ 4 pts

(b)

$[-2, \infty)$ 4 pts

(c)

$(-5, 0)$ 3 pts

(d)

$(0, 1)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -9^{+} : f(x) \to -2$ 2 pts

TEST PAGE 2

4. (a)

$2(x + 3)(x - 1)^2$ 16 pts

(b)

$x \to -\infty : f(x) \to -\infty$ 2 pts

(c)

$x \to \infty : f(x) \to \infty$ 2 pts

5. (a)

$(-1, -4)$ 6 pts

(b)

-4 4 pts

(c)

$x = -3, 1$ 10 pts

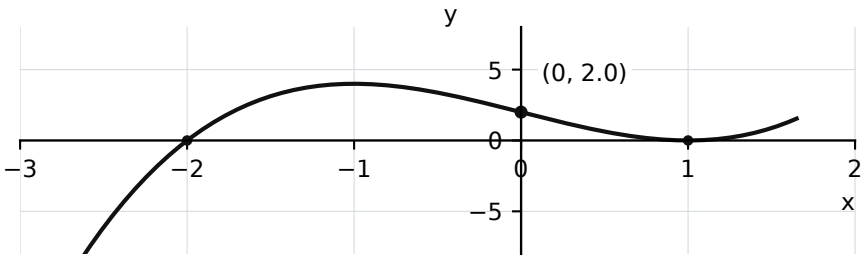
MATH 101 **56232**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	1. (a) 16 pts (b) 2 pts (c) 2 pts
2. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 2x^2 - 18$	2. (a) 6 pts (b) 4 pts (c) 10 pts
3. Find all zeros. $p(x) = x^3 - x^2 - 3x + 2$	3. 20 pts
4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-2)(x+4)}{(x-6)(x-2)}$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts

Test continues on the other side of the sheet.

MATH 101 **56232**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 2\sqrt{x+1} - 6$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

56232

TEST PAGE 1

1. (a)

$1(x+2)(x-1)^2$

16 pts

(b)

$x \to -\infty : f(x) \to -\infty$

2 pts

(c)

$x \to \infty : f(x) \to \infty$

2 pts

2. (a)

$(0, -18)$ 6 pts

(b)

-18 4 pts

(c)

$x = -3, 3$ 10 pts

3.

$x = 2, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 20 pts

4. (a)

$\mathbb{R} \setminus \{2, 6\}$ 4 pts

(b)

$(2, -\frac{3}{2})$ 4 pts

(c)

$x = 6$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-4, 0)$ 3 pts

(f)

$(0, -\frac{2}{3})$ 3 pts

TEST PAGE 2

5. (a)

$[-1, \infty)$ 4 pts

(b)

$[-6, \infty)$ 4 pts

(c)

$(8, 0)$ 3 pts

(d)

$(0, -4)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -1^+ : f(x) \to -6$ 2 pts

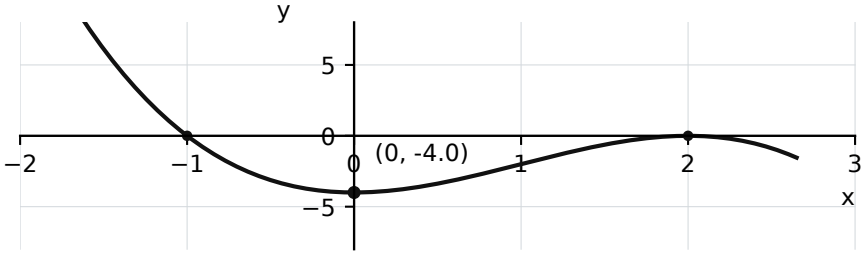
MATH 101 **83606**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-2)(x+4)}{(x-6)(x-2)}$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	2. (a) 16 pts (b) 2 pts (c) 2 pts
3. Find all zeros. $p(x) = x^3 + 4x^2 + 4x + 1$	3. 20 pts

Test continues on the other side of the sheet.

MATH 101 **83606**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 2\sqrt{x+4} - 6$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
5. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 2x^2 - 18$	5. (a) 6 pts (b) 4 pts (c) 10 pts

83606

TEST PAGE 1

1. (a)
$\mathbb{R} \setminus \{2, 6\}$ 4 pts

(b)
$(2, -\frac{3}{2})$ 4 pts

(c)
$x = 6$ 3 pts

(d)
$y = 1$ 3 pts

(e)
$(-4, 0)$ 3 pts

(f)
$(0, -\frac{2}{3})$ 3 pts

2. (a)
$-1(x+1)(x-2)^2$ 16 pts

(b)
$x \to -\infty : f(x) \to \infty$ 2 pts

(c)
$x \to \infty : f(x) \to -\infty$ 2 pts

3.
$x = -1, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 20 pts

TEST PAGE 2

4. (a)
$[-4, \infty)$ 4 pts

(b)
$[-6, \infty)$ 4 pts

(c)
$(5, 0)$ 3 pts

(d)
$(0, -2)$ 3 pts

(e)
$x \to \infty : f(x) \to \infty$ 4 pts

(f)
$x \to -4^+ : f(x) \to -6$ 2 pts

5. (a)
$(0, -18)$ 6 pts

(b)
-18 4 pts

(c)
$x = -3, 3$ 10 pts

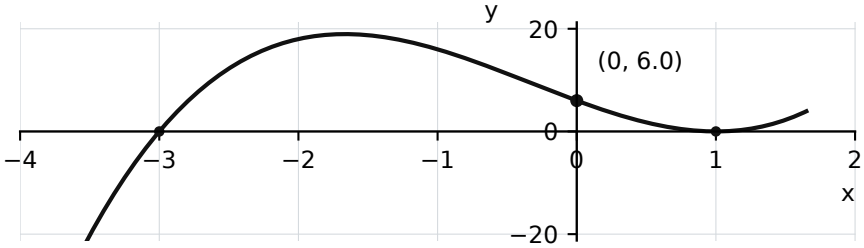
MATH 101 **49407**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-2)(x+4)}{(x-6)(x-2)}$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	2. (a) 16 pts (b) 2 pts (c) 2 pts
3. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 2x^2 - 18$	3. (a) 6 pts (b) 4 pts (c) 10 pts
4. Find all zeros. $p(x) = x^3 - 4x + 3$	4. 20 pts

Test continues on the other side of the sheet.

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 2\sqrt{x+9} - 6$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

49407

TEST PAGE 1

1. (a)
$\mathbb{R} \setminus \{2, 6\}$ 4 pts

(b)
$(2, -\frac{3}{2})$ 4 pts

(c)
$x = 6$ 3 pts

(d)
$y = 1$ 3 pts

(e)
$(-4, 0)$ 3 pts

(f)
$(0, -\frac{2}{3})$ 3 pts

2. (a)
$2(x+3)(x-1)^2$ 16 pts

(b)
$x \to -\infty : f(x) \to -\infty$ 2 pts

(c)
$x \to \infty : f(x) \to \infty$ 2 pts

3. (a)
$(0, -18)$ 6 pts

(b)
-18 4 pts

(c)
$x = -3, 3$ 10 pts

4.
$x = 1, -\frac{1}{2} + \frac{\sqrt{13}}{2}, -\frac{\sqrt{13}}{2} - \frac{1}{2}$ 20 pts

TEST PAGE 2

5. (a)
$[-9, \infty)$ 4 pts

(b)
$[-6, \infty)$ 4 pts

(c)
$(0, 0)$ 3 pts

(d)
$(0, 0)$ 3 pts

(e)
$x \to \infty : f(x) \to \infty$ 4 pts

(f)
$x \to -9^+ : f(x) \to -6$ 2 pts

MATH 101 **92695**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 3\sqrt{x+1} - 12$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
2. Find all zeros. $p(x) = x^3 - x^2 - 3x + 2$	2. 20 pts
3. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 3(x-1)^2 - 48$	3. (a) 6 pts (b) 4 pts (c) 10 pts

Test continues on the other side of the sheet.

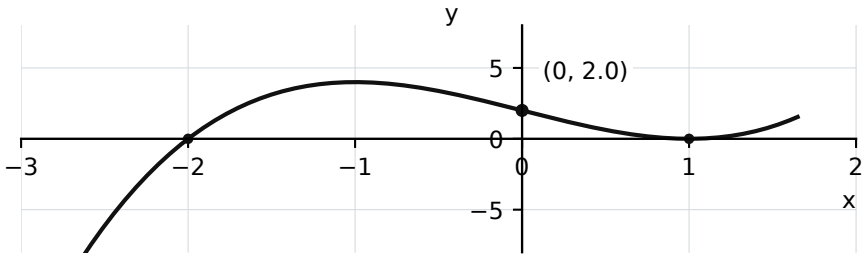
MATH 101 **92695**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-3)(x+5)}{(x-7)(x-3)}$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
5. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	5. (a) 16 pts (b) 2 pts (c) 2 pts

92695

TEST PAGE 1

1. (a)
$[-1, \infty)$ — 4 pts

(b)
$[-12, \infty)$ — 4 pts

(c)
$(15, 0)$ — 3 pts

(d)
$(0, -9)$ — 3 pts

(e)
$x \to \infty : f(x) \to \infty$ — 4 pts

(f)
$x \to -1^+ : f(x) \to -12$ — 2 pts

2.
$x = 2, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ — 20 pts

3. (a)
$(1, -48)$ — 6 pts

(b)
-48 — 4 pts

(c)
$x = -3, 5$ — 10 pts

TEST PAGE 2

4. (a)
$\mathbb{R} \setminus \{3, 7\}$ — 4 pts

(b)
$(3, -2)$ — 4 pts

(c)
$x = 7$ — 3 pts

(d)
$y = 1$ — 3 pts

(e)
$(-5, 0)$ — 3 pts

(f)
$(0, -\frac{5}{7})$ — 3 pts

5. (a)
$1(x + 2)(x - 1)^2$ — 16 pts

(b)
$x \to -\infty : f(x) \to -\infty$ — 2 pts

(c)
$x \to \infty : f(x) \to \infty$ — 2 pts

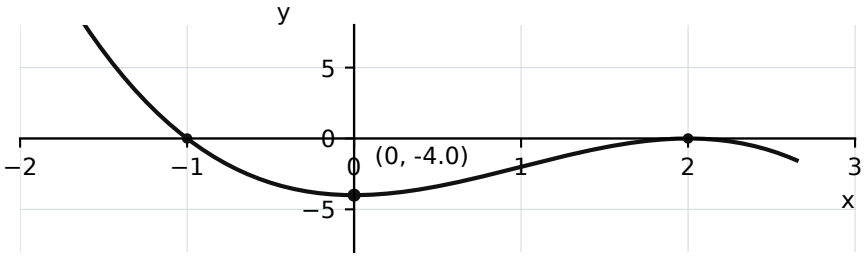
MATH 101 **42571**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 3\sqrt{x+4} - 12$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
2. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	2. (a) 16 pts (b) 2 pts (c) 2 pts
3. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 3(x-1)^2 - 48$	3. (a) 6 pts (b) 4 pts (c) 10 pts
4. Find all zeros. $p(x) = x^3 + 4x^2 + 4x + 1$	4. 20 pts

Test continues on the other side of the sheet.

MATH 101 **42571**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-3)(x+5)}{(x-7)(x-3)}$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts

42571

TEST PAGE 1

1. (a)
$[-4, \infty)$ 4 pts

(b)
$[-12, \infty)$ 4 pts

(c)
$(12, 0)$ 3 pts

(d)
$(0, -6)$ 3 pts

(e)
$x \to \infty : f(x) \to \infty$ 4 pts

(f)
$x \to -4^+ : f(x) \to -12$ 2 pts

2. (a)
$-1(x+1)(x-2)^2$
16 pts

(b)
$x \to -\infty : f(x) \to \infty$
2 pts

(c)
$x \to \infty : f(x) \to -\infty$
2 pts

3. (a)
$(1, -48)$ 6 pts

(b)
-48 4 pts

(c)
$x = -3, 5$ 10 pts

4.
$x = -1, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 20 pts

TEST PAGE 2

5. (a)
$\mathbb{R} \setminus \{3, 7\}$ 4 pts

(b)
$(3, -2)$ 4 pts

(c)
$x = 7$ 3 pts

(d)
$y = 1$ 3 pts

(e)
$(-5, 0)$ 3 pts

(f)
$(0, -\frac{5}{7})$ 3 pts

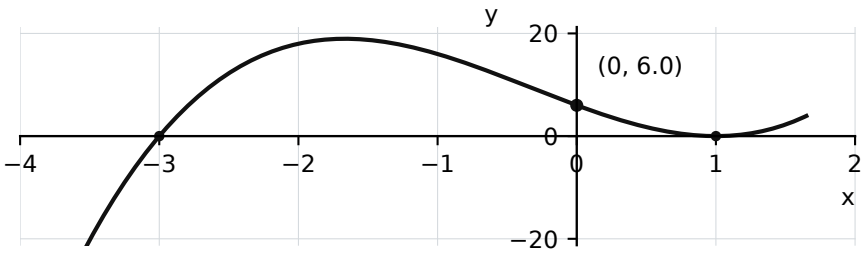
MATH 101 **32153**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end.  	1. (a) 16 pts (b) 2 pts (c) 2 pts
2. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-3)(x+5)}{(x-7)(x-3)}$	2. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
3. Find all zeros. $p(x) = x^3 - 4x + 3$	3. 20 pts

Test continues on the other side of the sheet.

MATH 101 **32153**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 3\sqrt{x+9} - 12$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
5. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 3(x-1)^2 - 48$	5. (a) 6 pts (b) 4 pts (c) 10 pts

32153

TEST PAGE 1

1. (a)

$2(x+3)(x-1)^2$

16 pts

(b)

$x \to -\infty : f(x) \to -\infty$

2 pts

(c)

$x \to \infty : f(x) \to \infty$

2 pts

2. (a)

$\mathbb{R} \setminus \{3, 7\}$ 4 pts

(b)

$(3, -2)$ 4 pts

(c)

$x = 7$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-5, 0)$ 3 pts

(f)

$(0, -\frac{5}{7})$ 3 pts

3.

$x = 1, -\frac{1}{2} + \frac{\sqrt{13}}{2}, -\frac{\sqrt{13}}{2} - \frac{1}{2}$ 20 pts

TEST PAGE 2

4. (a)

$[-9, \infty)$ 4 pts

(b)

$[-12, \infty)$ 4 pts

(c)

$(7, 0)$ 3 pts

(d)

$(0, -3)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -9^+ : f(x) \to -12$ 2 pts

5. (a)

$(1, -48)$ 6 pts

(b)

-48 4 pts

(c)

$x = -3, 5$ 10 pts

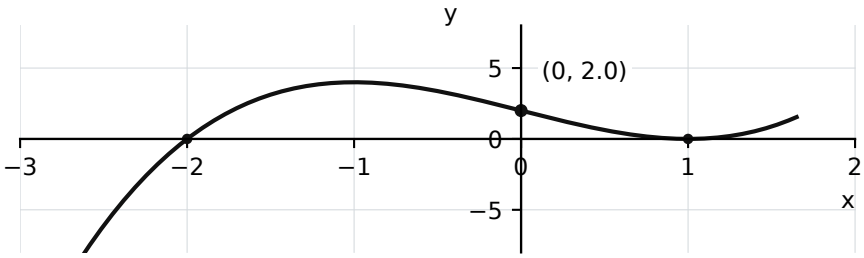
MATH 101 **51813**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 - x^2 - 3x + 2$	1. 20 pts
2. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	2. (a) 16 pts (b) 2 pts (c) 2 pts
3. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 4\sqrt{x+1} - 20$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
4. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 4(x-2)^2 - 100$	4. (a) 6 pts (b) 4 pts (c) 10 pts

Test continues on the other side of the sheet.

MATH 101 **51813**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-4)(x+6)}{(x-8)(x-4)}$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts

51813

TEST PAGE 1

1.

$x = 2, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 20 pts

2. (a)

$1(x+2)(x-1)^2$

16 pts

(b)

$x \to -\infty : f(x) \to -\infty$

2 pts

(c)

$x \to \infty : f(x) \to \infty$

2 pts

3. (a)

$[-1, \infty)$ 4 pts

(b)

$[-20, \infty)$ 4 pts

(c)

$(24, 0)$ 3 pts

(d)

$(0, -16)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -1^+ : f(x) \to -20$ 2 pts

4. (a)

$(2, -100)$ 6 pts

(b)

-100 4 pts

(c)

$x = -3, 7$ 10 pts

TEST PAGE 2

5. (a)

$\mathbb{R} \setminus \{4, 8\}$ 4 pts

(b)

$(4, -\frac{5}{2})$ 4 pts

(c)

$x = 8$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-6, 0)$ 3 pts

(f)

$(0, -\frac{3}{4})$ 3 pts

MATH 101 **15899**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 4\sqrt{x+4} - 20$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
2. Find all zeros. $p(x) = x^3 + 4x^2 + 4x + 1$	2. 20 pts
3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-4)(x+6)}{(x-8)(x-4)}$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts

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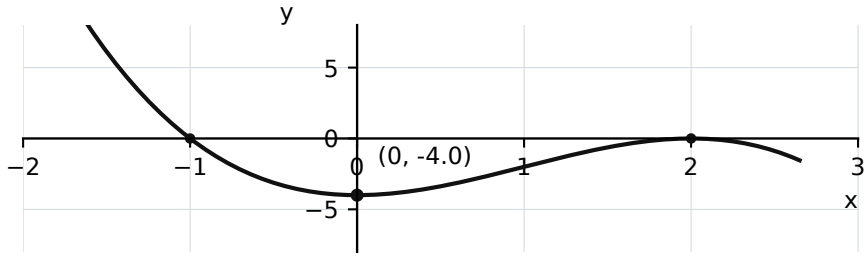
MATH 101 **15899**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 4(x-2)^2 - 100$	4. (a) 6 pts (b) 4 pts (c) 10 pts
5. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	5. (a) 16 pts (b) 2 pts (c) 2 pts

15899

TEST PAGE 1

1. (a)
$[-4, \infty)$ 4 pts

(b)
$[-20, \infty)$ 4 pts

(c)
$(21, 0)$ 3 pts

(d)
$(0, -12)$ 3 pts

(e)
$x \to \infty : f(x) \to \infty$ 4 pts

(f)
$x \to -4^+ : f(x) \to -20$ 2 pts

2.
$x = -1, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 20 pts

3. (a)
$\mathbb{R} \setminus \{4, 8\}$ 4 pts

(b)
$(4, -\frac{5}{2})$ 4 pts

(c)
$x = 8$ 3 pts

(d)
$y = 1$ 3 pts

(e)
$(-6, 0)$ 3 pts

(f)
$(0, -\frac{3}{4})$ 3 pts

TEST PAGE 2

4. (a)
$(2, -100)$ 6 pts

(b)
-100 4 pts

(c)
$x = -3, 7$ 10 pts

5. (a)
$-1(x+1)(x-2)^2$ 16 pts

(b)
$x \to -\infty : f(x) \to \infty$ 2 pts

(c)
$x \to \infty : f(x) \to -\infty$ 2 pts

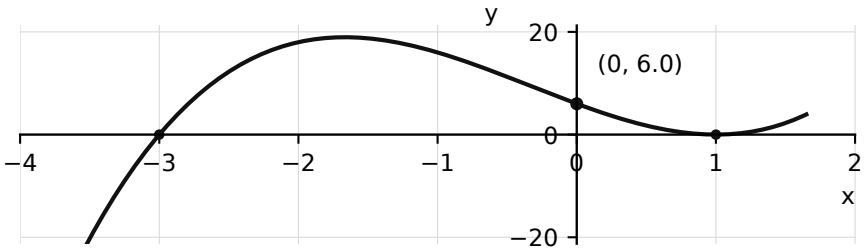
MATH 101 **23988**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 - 4x + 3$	1. 20 pts
2. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 4(x-2)^2 - 100$	2. (a) 6 pts (b) 4 pts (c) 10 pts
3. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	3. (a) 16 pts (b) 2 pts (c) 2 pts
4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-4)(x+6)}{(x-8)(x-4)}$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts

Test continues on the other side of the sheet.

MATH 101 **23988**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 4\sqrt{x+9} - 20$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

23988

TEST PAGE 1

1.

$x = 1,\ -\frac{1}{2} + \frac{\sqrt{13}}{2},\ -\frac{\sqrt{13}}{2} - \frac{1}{2}$ 20 pts

2. (a)

$(2, -100)$ 6 pts

(b)

-100 4 pts

(c)

$x = -3, 7$ 10 pts

3. (a)

$2(x+3)(x-1)^2$ 16 pts

(b)

$x \to -\infty : f(x) \to -\infty$ 2 pts

(c)

$x \to \infty : f(x) \to \infty$ 2 pts

4. (a)

$\mathbb{R} \setminus \{4, 8\}$ 4 pts

(b)

$(4, -\frac{5}{2})$ 4 pts

(c)

$x = 8$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-6, 0)$ 3 pts

(f)

$(0, -\frac{3}{4})$ 3 pts

TEST PAGE 2

5. (a)

$[-9, \infty)$ 4 pts

(b)

$[-20, \infty)$ 4 pts

(c)

$(16, 0)$ 3 pts

(d)

$(0, -8)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -9^+ : f(x) \to -20$ 2 pts

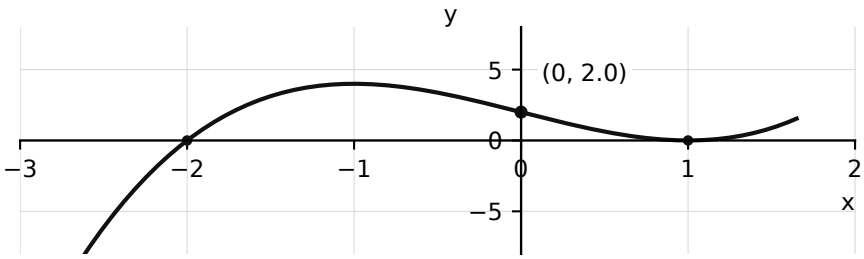
MATH 101 **85069**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	1. (a) 16 pts (b) 2 pts (c) 2 pts
2. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 5(x-3)^2 - 180$	2. (a) 6 pts (b) 4 pts (c) 10 pts
3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-5)(x+7)}{(x-9)(x-5)}$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
4. Find all zeros. $p(x) = x^3 - x^2 - 3x + 2$	4. 20 pts

Test continues on the other side of the sheet.

MATH 101 **85069**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 5\sqrt{x+1} - 30$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

85069

TEST PAGE 1

1. (a)

$1(x+2)(x-1)^2$

16 pts

(b)

$x \to -\infty : f(x) \to -\infty$

2 pts

(c)

$x \to \infty : f(x) \to \infty$

2 pts

2. (a)

$(3, -180)$ 6 pts

(b)

-180 4 pts

(c)

$x = -3, 9$ 10 pts

3. (a)

$\mathbb{R} \setminus \{5, 9\}$ 4 pts

(b)

$(5, -3)$ 4 pts

(c)

$x = 9$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-7, 0)$ 3 pts

(f)

$(0, -\frac{7}{9})$ 3 pts

4.

$x = 2, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 20 pts

TEST PAGE 2

5. (a)

$[-1, \infty)$ 4 pts

(b)

$[-30, \infty)$ 4 pts

(c)

$(35, 0)$ 3 pts

(d)

$(0, -25)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -1^+ : f(x) \to -30$ 2 pts

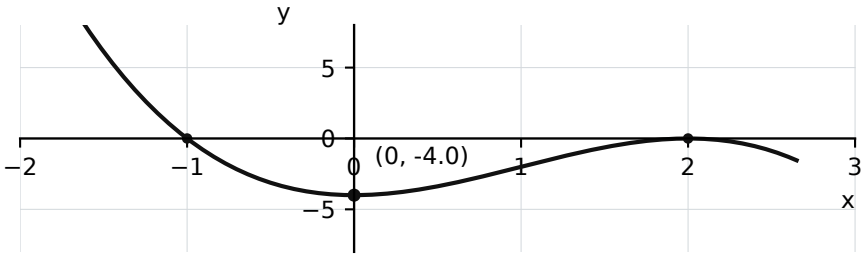
MATH 101 **91724**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	1. (a) 16 pts (b) 2 pts (c) 2 pts
2. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 5(x-3)^2 - 180$	2. (a) 6 pts (b) 4 pts (c) 10 pts
3. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 5\sqrt{x+4} - 30$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **91724**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-5)(x+7)}{(x-9)(x-5)}$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
5. Find all zeros. $p(x) = x^3 + 4x^2 + 4x + 1$	5. 20 pts

91724

TEST PAGE 1

1. (a)

$-1(x+1)(x-2)^2$

16 pts

(b)

$x \to -\infty : f(x) \to \infty$

2 pts

(c)

$x \to \infty : f(x) \to -\infty$

2 pts

2. (a)

$(3, -180)$ 6 pts

(b)

-180 4 pts

(c)

$x = -3, 9$ 10 pts

3. (a)

$[-4, \infty)$ 4 pts

(b)

$[-30, \infty)$ 4 pts

(c)

$(32, 0)$ 3 pts

(d)

$(0, -20)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -4^+ : f(x) \to -30$ 2 pts

TEST PAGE 2

4. (a)

$\mathbb{R} \setminus \{5, 9\}$ 4 pts

(b)

$(5, -3)$ 4 pts

(c)

$x = 9$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-7, 0)$ 3 pts

(f)

$(0, -\frac{7}{9})$ 3 pts

5.

$x = -1, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 20 pts

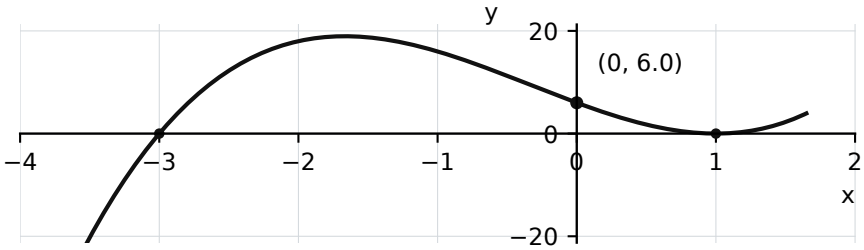
MATH 101 **27180**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 - 4x + 3$	1. 20 pts
2. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-5)(x+7)}{(x-9)(x-5)}$	2. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
3. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	3. (a) 16 pts (b) 2 pts (c) 2 pts
4. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 5(x-3)^2 - 180$	4. (a) 6 pts (b) 4 pts (c) 10 pts

Test continues on the other side of the sheet.

MATH 101 **27180**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 5\sqrt{x+9} - 30$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

27180

TEST PAGE 1

1.

$x = 1,\ -\frac{1}{2} + \frac{\sqrt{13}}{2},\ -\frac{\sqrt{13}}{2} - \frac{1}{2}$ 20 pts

2. (a)

$\mathbb{R} \setminus \{5, 9\}$ 4 pts

(b)

$(5,\ -3)$ 4 pts

(c)

$x = 9$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-7, 0)$ 3 pts

(f)

$(0,\ -\frac{7}{9})$ 3 pts

3. (a)

$2(x+3)(x-1)^2$

16 pts

(b)

$x \to -\infty : f(x) \to -\infty$

2 pts

(c)

$x \to \infty : f(x) \to \infty$

2 pts

4. (a)

$(3,\ -180)$ 6 pts

(b)

-180 4 pts

(c)

$x = -3, 9$ 10 pts

TEST PAGE 2

5. (a)

$[-9, \infty)$ 4 pts

(b)

$[-30, \infty)$ 4 pts

(c)

$(27, 0)$ 3 pts

(d)

$(0,\ -15)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -9^{+} : f(x) \to -30$ 2 pts

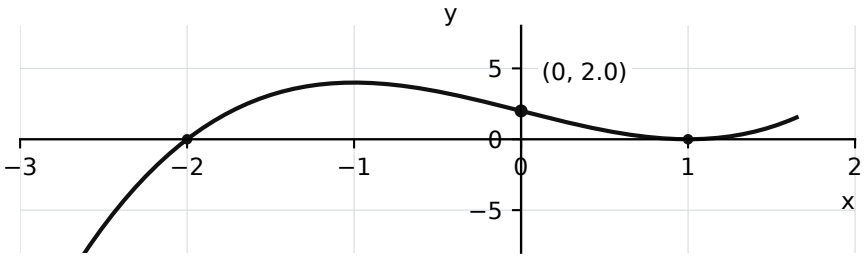
MATH 101 **76796**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-6)(x+8)}{(x-10)(x-6)}$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Find all zeros. $p(x) = x^3 - x^2 - 3x + 2$	2. 20 pts
3. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 6(x-4)^2 - 294$	3. (a) 6 pts (b) 4 pts (c) 10 pts
4. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	4. (a) 16 pts (b) 2 pts (c) 2 pts

Test continues on the other side of the sheet.

MATH 101 **76796**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 6\sqrt{x+1} - 42$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts

76796

TEST PAGE 1

1. (a)
$\mathbb{R} \setminus \{6, 10\}$ 4 pts

(b)
$(6, -\frac{7}{2})$ 4 pts

(c)
$x = 10$ 3 pts

(d)
$y = 1$ 3 pts

(e)
$(-8, 0)$ 3 pts

(f)
$(0, -\frac{4}{5})$ 3 pts

2.
$x = 2, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 20 pts

3. (a)
$(4, -294)$ 6 pts

(b)
-294 4 pts

(c)
$x = -3, 11$ 10 pts

4. (a)
$1(x+2)(x-1)^2$
16 pts

(b)
$x \to -\infty : f(x) \to -\infty$
2 pts

(c)
$x \to \infty : f(x) \to \infty$
2 pts

TEST PAGE 2

5. (a)
$[-1, \infty)$ 4 pts

(b)
$[-42, \infty)$ 4 pts

(c)
$(48, 0)$ 3 pts

(d)
$(0, -36)$ 3 pts

(e)
$x \to \infty : f(x) \to \infty$ 4 pts

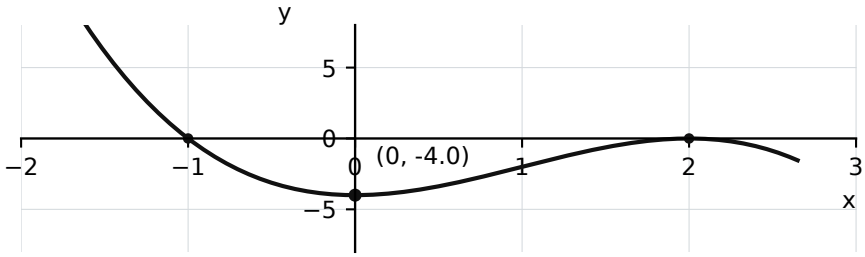
(f)
$x \to -1^+ : f(x) \to -42$ 2 pts

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 6\sqrt{x+4} - 42$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
2. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	2. (a) 16 pts (b) 2 pts (c) 2 pts

Test continues on the other side of the sheet.

MATH 101 **36361**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-6)(x+8)}{(x-10)(x-6)}$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
4. Find all zeros. $p(x) = x^3 + 4x^2 + 4x + 1$	4. 20 pts
5. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 6(x-4)^2 - 294$	5. (a) 6 pts (b) 4 pts (c) 10 pts

36361

TEST PAGE 1

1. (a)
$[-4, \infty)$ 4 pts

(b)
$[-42, \infty)$ 4 pts

(c)
$(45, 0)$ 3 pts

(d)
$(0, -30)$ 3 pts

(e)
$x \to \infty : f(x) \to \infty$ 4 pts

(f)
$x \to -4^+ : f(x) \to -42$ 2 pts

2. (a)
$-1(x+1)(x-2)^2$ 16 pts

(b)
$x \to -\infty : f(x) \to \infty$ 2 pts

(c)
$x \to \infty : f(x) \to -\infty$ 2 pts

TEST PAGE 2

3. (a)
$\mathbb{R} \setminus \{6, 10\}$ 4 pts

(b)
$(6, -\frac{7}{2})$ 4 pts

(c)
$x = 10$ 3 pts

(d)
$y = 1$ 3 pts

(e)
$(-8, 0)$ 3 pts

(f)
$(0, -\frac{4}{5})$ 3 pts

4.
$x = -1, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 20 pts

5. (a)
$(4, -294)$ 6 pts

(b)
-294 4 pts

(c)
$x = -3, 11$ 10 pts

MATH 101 **93632**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 - 4x + 3$	1. 20 pts
2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 6\sqrt{x+9} - 42$	2. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
3. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 6(x-4)^2 - 294$	3. (a) 6 pts (b) 4 pts (c) 10 pts

Test continues on the other side of the sheet.

MATH 101 **93632**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept.

$$f(x) = \frac{(x-6)(x+8)}{(x-10)(x-6)}$$

4. (a) 4 pts

(b) 4 pts

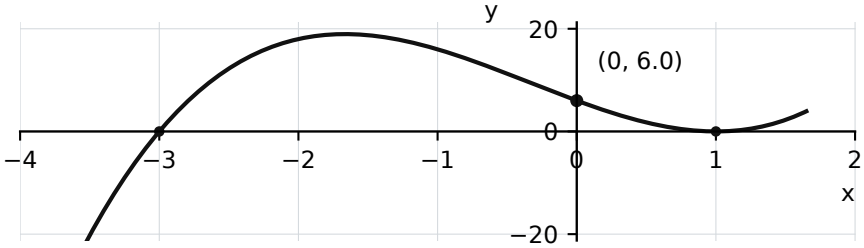
(c) 3 pts

(d) 3 pts

(e) 3 pts

(f) 3 pts

5. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end.



5. (a) 16 pts

(b) 2 pts

(c) 2 pts

93632

TEST PAGE 1

1.

$x = 1,\ -\frac{1}{2} + \frac{\sqrt{13}}{2},\ -\frac{\sqrt{13}}{2} - \frac{1}{2}$ 20 pts

2. (a)

$[-9, \infty)$ 4 pts

(b)

$[-42, \infty)$ 4 pts

(c)

$(40, 0)$ 3 pts

(d)

$(0,\ -24)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -9^{+} : f(x) \to -42$ 2 pts

3. (a)

$(4,\ -294)$ 6 pts

(b)

-294 4 pts

(c)

$x = -3, 11$ 10 pts

TEST PAGE 2

4. (a)

$\mathbb{R} \setminus \{6, 10\}$ 4 pts

(b)

$(6,\ -\frac{7}{2})$ 4 pts

(c)

$x = 10$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-8, 0)$ 3 pts

(f)

$(0,\ -\frac{4}{5})$ 3 pts

5. (a)

$2(x+3)(x-1)^2$

16 pts

(b)

$x \to -\infty : f(x) \to -\infty$

2 pts

(c)

$x \to \infty : f(x) \to \infty$

2 pts

MATH 101 **48388**

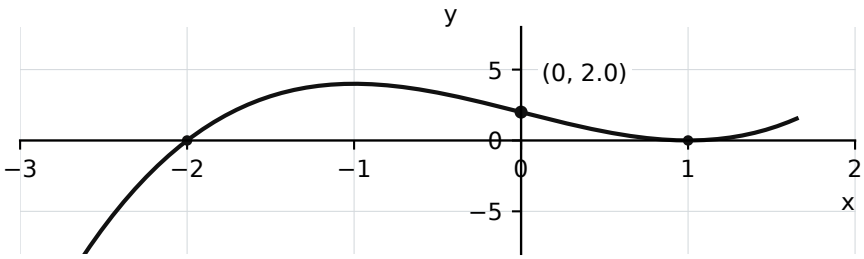
Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end.



1. (a)

16 pts

(b)

2 pts

(c)

2 pts

2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$f(x) = 7\sqrt{x+1} - 56$

2. (a)

4 pts

(b)

4 pts

(c)

3 pts

(d)

3 pts

(e)

4 pts

(f)

2 pts

Test continues on the other side of the sheet.

MATH 101 **48388**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-7)(x+9)}{(x-11)(x-7)}$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
4. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 7(x-5)^2 - 448$	4. (a) 6 pts (b) 4 pts (c) 10 pts
5. Find all zeros. $p(x) = x^3 - x^2 - 3x + 2$	5. 20 pts

48388

TEST PAGE 1

1. (a)

$1(x+2)(x-1)^2$

16 pts

(b)

$x \to -\infty : f(x) \to -\infty$

2 pts

(c)

$x \to \infty : f(x) \to \infty$

2 pts

2. (a)

$[-1, \infty)$ 4 pts

(b)

$[-56, \infty)$ 4 pts

(c)

$(63, 0)$ 3 pts

(d)

$(0, -49)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -1^+ : f(x) \to -56$ 2 pts

TEST PAGE 2

3. (a)

$\mathbb{R} \setminus \{7, 11\}$ 4 pts

(b)

$(7, -4)$ 4 pts

(c)

$x = 11$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-9, 0)$ 3 pts

(f)

$(0, -\frac{9}{11})$ 3 pts

4. (a)

$(5, -448)$ 6 pts

(b)

-448 4 pts

(c)

$x = -3, 13$ 10 pts

5.

$x = 2, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 20 pts

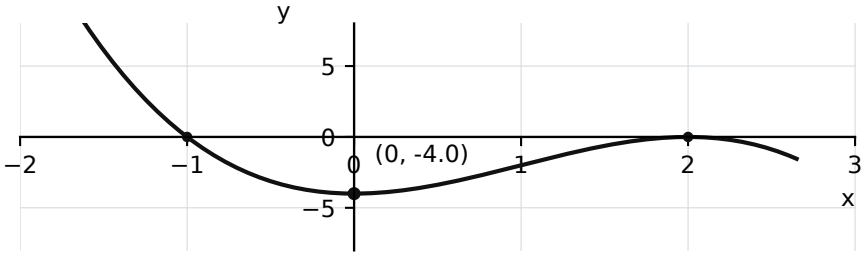
MATH 101 **59681**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 + 4x^2 + 4x + 1$	1. 20 pts
2. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. Record: (a) polynomial; (b) left end; (c) right end. 	2. (a) 16 pts (b) 2 pts (c) 2 pts
3. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 7\sqrt{x+4} - 56$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 4 pts (f) 2 pts
4. Find the vertex and the minimum or maximum value (state which); all zeros. Record: (a) vertex; (b) minimum; (c) zeros. $f(x) = 7(x-5)^2 - 448$	4. (a) 6 pts (b) 4 pts (c) 10 pts

Test continues on the other side of the sheet.

MATH 101 **59681**

Unit 3: Polynomial, Rational, and Radical Functions

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x-7)(x+9)}{(x-11)(x-7)}$	5. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts

59681

TEST PAGE 1

1.

$x = -1,\ -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 20 pts

2. (a)

$-1(x+1)(x-2)^2$ 16 pts

(b)

$x \to -\infty : f(x) \to \infty$ 2 pts

(c)

$x \to \infty : f(x) \to -\infty$ 2 pts

3. (a)

$[-4, \infty)$ 4 pts

(b)

$[-56, \infty)$ 4 pts

(c)

$(60, 0)$ 3 pts

(d)

$(0,\ -42)$ 3 pts

(e)

$x \to \infty : f(x) \to \infty$ 4 pts

(f)

$x \to -4^+ : f(x) \to -56$ 2 pts

4. (a)

$(5,\ -448)$ 6 pts

(b)

-448 4 pts

(c)

$x = -3, 13$ 10 pts

TEST PAGE 2

5. (a)

$\mathbb{R} \setminus \{7, 11\}$ 4 pts

(b)

$(7,\ -4)$ 4 pts

(c)

$x = 11$ 3 pts

(d)

$y = 1$ 3 pts

(e)

$(-9, 0)$ 3 pts

(f)

$(0,\ -\frac{9}{11})$ 3 pts